ModelIt!

Student Activity Pack

When the Earth Cracks Open

The Powerful Forces That Shape Our Planet's Surface

Grade 6 | NGSS MS-ESS2-2 | Stability and Change

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Have you ever felt an earthquake or seen one on the news? What happened?

2 Why do you think earthquakes happen in some places but not others?

3 What do you think is beneath the ground under your feet?

4 Draw what you think the inside of the Earth looks like.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Plate Movement Speed | Mantle Heat | Rock Density | Boundary Friction

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Slow Drift

Low mantle heat, slow plate movement

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Heat Surge

Lock Mantle Heat to maximum

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Friction Lockup

Set Boundary Friction to maximum

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Magma Pressure — The buildup of molten rock beneath the surface that can cause volcanic eruptions

Water Content — Water in rock lowers its melting point, affecting where magma forms

Plate Thickness — Thicker plates resist bending while thinner plates are more flexible

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why do earthquakes and volcanoes happen in the same places over and over again? Use evidence from your simulation to support your answers.

1 Why do earthquakes and volcanoes form a 'Ring of Fire' around the Pacific Ocean?

2 How does friction at a plate boundary actually CAUSE bigger earthquakes, not smaller ones?

3 If tectonic plates move so slowly (cm/year), how can they cause such violent events?

4 What would Earth's surface look like if there were no tectonic plates?

5 How does your model help predict where future earthquakes might occur?

8 STEM Challenge: Design an Earthquake-Ready City

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YOUR ENGINEERING MISSION

Design a city near a plate boundary that can withstand earthquakes using evidence from your tectonic model about earthquake intensity and frequency.

THE SCENARIO

A new city is being planned near the San Andreas Fault in California. The city council needs your engineering team to recommend building designs, infrastructure, and emergency systems based on what your model reveals about earthquake patterns.

Engineering Design Thinking

1. WHAT BUILDING MATERIALS AND FOUNDATIONS WILL RESIST EARTHQUAKE SHAKING?
2. WHERE SHOULD EMERGENCY SHELTERS AND HOSPITALS BE LOCATED RELATIVE TO THE FAULT?
3. WHAT EARLY WARNING SYSTEMS WILL YOU INCLUDE IN YOUR CITY DESIGN?
4. HOW WILL YOUR CITY'S INFRASTRUCTURE (ROADS, BRIDGES, WATER PIPES) HANDLE GROUND MOVEMENT?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Seismologists and Geophysicists study earthquakes and Earth's interior to predict hazards and design safer communities. They earn $70,000–$130,000/year.